

NEW CASTLE, DE, USA

WMO: 724180

Lat: 39.674N

Lon: 75.606W

Elev: 79

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13781

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	16.9	-3.4	4.6	15.8	0.9	5.8	19.7	28.7	32.0	26.6	31.9	10.6	290	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	92.1	75.2	89.5	74.1	87.0	73.2	78.2	87.3	76.9	85.2	75.7	83.1	10.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	134.2	81.9	74.4	129.0	80.8	73.3	124.0	79.7	41.6	87.6	40.2	85.3	39.1	83.2	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	21.1	18.7	7.4	96.1	4.8	2.9	4.0	98.2	1.2	99.9	-1.5	101.6	-5.0	103.7
			5.7	80.8	4.7	1.7	2.3	82.0	-0.5	83.0	-3.1	83.9	-6.5	85.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	33.0	35.3	43.0	54.0	63.2	72.5	77.2	75.5	69.0	57.3	46.5	38.2
	DBStd	17.23	9.91	9.11	9.01	8.15	7.47	6.14	4.90	4.83	6.48	7.93	7.98	8.58
	HDD50	1814	533	418	255	46	1	0	0	0	21	160	380	
	HDD65	4677	992	831	683	344	127	12	0	1	34	264	557	832
	CDD50	3815	6	7	38	164	411	674	842	791	569	246	54	13
	CDD65	1202	0	0	1	13	72	236	377	326	152	25	0	0
	CDH74	10203	0	0	16	143	596	1969	3559	2743	1037	138	2	0
	CDH80	3410	0	0	2	34	165	638	1376	899	273	23	0	0
Wind	WSAvg	8.5	9.3	9.6	10.2	9.8	8.3	7.8	7.2	6.9	7.5	7.8	8.4	8.7
Precipitation	PrecAvg	41.3	3.0	2.8	3.8	3.3	3.9	3.7	4.1	3.4	4.0	2.8	3.0	3.4
	PrecMax	55.3	8.4	7.0	9.2	6.8	7.4	9.9	12.6	11.2	12.7	6.4	7.8	7.9
	PrecMin	24.9	0.5	0.4	0.8	0.4	0.2	0.2	0.9	0.2	0.8	0.1	0.5	0.8
	PrecStd	7.3	1.6	1.5	1.9	1.6	1.8	1.9	2.3	2.1	2.6	1.5	1.9	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.6	67.4	75.6	84.4	89.1	93.3	96.4	94.1	90.9	82.8	71.4	67.0
		MCWB	57.6	57.6	60.8	65.7	72.2	75.4	77.3	76.1	73.0	70.0	62.2	61.4
	2%	DB	56.9	59.4	67.8	78.2	84.5	89.6	92.8	90.4	86.7	77.9	66.8	60.5
		MCWB	52.8	51.4	56.1	63.4	70.4	73.7	75.8	75.1	72.0	67.4	59.3	56.1
	5%	DB	51.9	53.0	62.2	72.9	80.8	86.4	90.0	87.5	82.8	73.7	63.1	55.4
		MCWB	47.9	45.9	52.3	60.4	68.1	72.5	74.8	73.8	70.6	65.1	56.9	51.5
	10%	DB	46.9	48.5	56.8	68.1	76.6	83.6	87.4	84.9	80.0	70.4	59.4	51.1
		MCWB	42.9	42.7	49.3	57.5	65.8	71.1	74.0	72.3	69.1	63.2	54.3	46.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	60.3	63.6	69.3	74.9	77.9	81.2	79.5	77.3	73.3	65.1	62.6
		MCDB	61.9	66.2	73.7	78.8	85.3	88.5	90.0	89.1	84.8	78.1	69.2	65.5
	2%	WB	53.9	52.8	58.5	65.5	72.4	76.2	78.7	77.8	75.5	70.4	61.8	57.1
		MCDB	55.6	57.0	65.0	74.7	81.5	85.3	88.1	86.1	81.1	75.4	64.8	60.0
	5%	WB	48.2	47.3	54.4	62.5	70.2	75.0	77.3	76.4	73.9	67.6	58.5	52.6
		MCDB	51.1	51.6	60.0	70.2	78.1	83.1	86.3	83.5	79.2	71.9	61.8	54.8
	10%	WB	43.6	43.3	50.4	59.5	67.9	73.5	76.0	75.2	72.1	64.3	55.3	47.8
		MCDB	46.6	48.1	56.0	66.6	74.6	80.5	83.9	81.5	77.2	68.7	58.5	50.3
Mean Daily Temperature Range	5% DB	MDBR	14.2	15.6	17.4	19.3	18.4	17.5	17.1	16.8	17.7	18.0	16.9	14.2
		MCDBR	20.5	22.9	25.3	25.7	22.7	20.2	19.8	18.8	20.2	21.1	21.3	19.5
		MCWBR	17.1	16.7	16.1	13.8	11.0	9.0	7.5	7.9	9.0	11.8	14.7	17.0
	5% WB	MCDBR	19.4	21.0	23.0	22.5	19.9	17.6	17.0	15.9	15.6	17.3	17.6	17.7
		MCWBR	17.8	17.5	17.1	14.1	10.7	8.9	7.3	7.6	8.5	11.8	14.5	17.3
Clear-Sky Solar Irradiance	taub	0.315	0.321	0.363	0.403	0.447	0.511	0.519	0.514	0.425	0.363	0.327	0.311	
	taud	2.445	2.417	2.361	2.276	2.186	2.048	2.039	2.039	2.264	2.443	2.503	2.503	
	Ebn at Noon	270	285	282	277	266	248	244	242	259	266	264	264	
	Edh at Noon	26	30	36	41	46	53	53	52	39	29	24	23	
All-Sky Solar Radiation	RadAvg	622	905	1190	1542	1725	1891	1907	1675	1380	1000	713	542	
	RadStd	53	102	94	128	211	142	87	102	153	104	62	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (79 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+239	+140

Nomenclature: See separate page